EXPERIMENT-1

Aim

To install Git, create a Git repository, and understand basic Git operations such as add, commit, modify, and view, along with transferring data between local and remote repositories.

Theory

Git is a distributed version control system used to track changes in files during software development. It helps developers manage different versions of a project, collaborate with others, and maintain a history of modifications made to the code.

A **Git repository** is a storage location where project files and their complete revision history are maintained. A repository can exist on a developer's local system or on a remote server such as **GitHub**.

A **local repository** is the repository stored on a developer's computer where files are created, modified, and committed. A **remote repository** is hosted on a server and allows developers to share their work with others.

Git uses several commands to manage files and track changes:

- **git init**
  This command initializes a new Git repository in a project folder. It creates a hidden .git directory that stores all version control information.

- **git add**
  This command adds files to the **staging area**, which prepares them to be included in the next commit.

- **git commit**
  The commit command saves the changes made in the staging area to the repository along with a descriptive message. Each commit represents a snapshot of the project at a specific time.

- **git status**
  This command displays the current state of the repository, showing modified files, staged files, and untracked files.

- **git log**
  This command shows the history of commits in the repository, including commit messages, authors, and timestamps.

Git also allows synchronization between **local and remote repositories**:

- **git remote add origin** is used to connect a local repository with a remote repository.
- **git push** uploads local commits to the remote repository.
- **git clone** copies a remote repository to a local machine.
- **git pull** downloads the latest updates from the remote repository to the local repository.

Using these commands, developers can efficiently track project changes, manage versions, and collaborate with other team members.

```
PS C:\Users\Chandrayan Paul\Documents\testing> git --version
git version 2.47.1.windows.1
PS C:\Users\Chandrayan Paul\Documents\testing> git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=main
```

```
PS C:\Users\Chandrayan Paul\Documents\testing>
PS C:\Users\Chandrayan Paul\Documents\testing> mkdir test


    Directory: C:\Users\Chandrayan Paul\Documents\testing


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        13-03-2026     22:50                test


PS C:\Users\Chandrayan Paul\Documents\testing> cd test
PS C:\Users\Chandrayan Paul\Documents\testing\test> git init
Initialized empty Git repository in C:/Users/Chandrayan Paul/Documents/testing/test/.git/
```

```
Initialized empty Git repository in C:/Users/Chandrayan Paul/Documents/testing/test/.git/
PS C:\Users\Chandrayan Paul\Documents\testing\test> git add .
PS C:\Users\Chandrayan Paul\Documents\testing\test> git add .
PS C:\Users\Chandrayan Paul\Documents\testing\test> git commit -m "Initial commit"
[main (root-commit) 8213aaa] Initial commit
 1 file changed, 11 insertions(+)
 create mode 100644 index.html
PS C:\Users\Chandrayan Paul\Documents\testing\test> git status
On branch main
nothing to commit, working tree clean
PS C:\Users\Chandrayan Paul\Documents\testing\test> git remote add origin https://github.com/Ron-Chandrayan/testing.gi
```

```
PS C:\Users\Chandrayan Paul\Documents\testing\test> git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 379 bytes | 47.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Ron-Chandrayan/testing.git
 * [new branch]      main -> main
PS C:\Users\Chandrayan Paul\Documents\testing\test> cd..
PS C:\Users\Chandrayan Paul\Documents\testing> git clone https://github.com/Ron-Chandrayan/forms.git
Cloning into 'forms'...
remote: Enumerating objects: 2084, done.
remote: Counting objects: 100% (2084/2084), done.
remote: Compressing objects: 100% (1753/1753), done.
remote: Total 2084 (delta 262), reused 2061 (delta 249), pack-reused 0 (from 0)
Receiving objects: 100% (2084/2084), 3.29 MiB | 2.45 MiB/s, done.
Resolving deltas: 100% (262/262), done.
Updating files: 100% (1844/1844), done.
```

```
PS C:\Users\Chandrayan Paul\Documents\testing> cd forms
PS C:\Users\Chandrayan Paul\Documents\testing\forms> git pull
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 971 bytes | 19.00 KiB/s, done.
From https://github.com/Ron-Chandrayan/forms
   00a9f22..230250d  main       -> origin/main
Updating 00a9f22..230250d
Fast-forward
 index.js | 4 ++--
 1 file changed, 2 insertions(+), 2 deletions(-)
```

```
PS C:\Users\Chandrayan Paul\Documents\testing\forms> git log
commit 230250d33db87b88a3048d584ce2450b67af55da (HEAD -> main, origin/main, origin/HEAD)
Author: Chandrayan Paul <chandrayanpaul27@gmail.com>
Date:   Fri Mar 13 22:56:38 2026 +0530

    Refactor index.js for clarity and structure

commit 00a9f22726990ef50b6c7409bcf186837cf4aca1
Author: Chandrayan Paul <chandrayanpaul27@gmail.com>
Date:   Wed Jan 29 20:45:14 2025 +0530

    final commit

commit 1bd0cf4ed88452e7457c94ba389a6bfcfa9f49ce
Author: Chandrayan Paul <chandrayanpaul27@gmail.com>
Date:   Wed Jan 29 20:42:09 2025 +0530

    final commit
```

Conclusion

This practical demonstrated the use of HTML5 semantic tags, CSS styling, and Bootstrap framework to design structured and responsive web pages. The experiment helped in understanding modern web development practices and building visually organized web interfaces.